Curated to Win: The Score Gap Bounds the Cost of Reordering Reward-Model Leaderboards

Anonymous ACL submission

Abstract

Automated research and self-improvement pipelines select reward models and LLM judges by their accuracy on a validation set, and that set is often curated by the pipeline or its designers. We measure how much item selection alone can change which evaluator wins. For any weighted-accuracy benchmark, we prove that a trailing model can overtake the leader only if the removed items carry at least the score gap in benchmark weight. We then solve a linear program over item weights, round each solution to an item subset and verify it, to find a removal that places each model first. On RewardBench (118 models, 2985 items) and RewardBench 2 (188 models, 1753 items), the removal needed has a median of 1.00× the gap, and at most 1.02× for RewardBench's top 20. Removing 19 items (0.7% of the score weight) makes RewardBench's runner-up the leader, and every top-20 model can be placed first while keeping 89.5% of the benchmark. On RewardBench, holding each section's share fixed raises the median cost to 1.39× the gap. Without any adversary, a random 100-item validation set keeps RewardBench's leader first in only 35% of draws. We recommend reporting a selection's curation margin and fixing validation items before any candidate is scored.

1 Introduction

Systems that automate research or improve themselves need an evaluator: a reward model to optimise against, or an LLM judge that ranks candidate outputs (Lu et al., 2024; Zheng et al., 2023). Evaluators are chosen by meta-evaluation, that is, by their agreement with human preference labels on a benchmark such as RewardBench (Lambert et al., 2024; Malik et al., 2025). In an automated pipeline, the items used for this choice are often selected by the pipeline itself: a filtered subset, a domain slice, or a small validation set. If the ranking of evaluators depends on which items are kept, whoever selects the items also selects the evaluator. This is reward hacking one level above the reward (Gao et al., 2022).

Leaderboards are known to be sensitive to benchmark choice, answer format and private testing (Dehghani et al., 2021; Alzahrani et al., 2024; Singh et al., 2025; Zhang and Hardt, 2024), and to targeted edits of individual items (Dubrovsky et al., 2025). Two recent analyses are closest to ours. Gordienko et al. (2026) count the *datasets* a developer must train on to top a multi-task leaderboard and show that the problem is NP-hard under mean win rate. Oyarhoseini et al. (2026) use influence approximations to show that sub-1% edits to pairwise matches can change the leader of a Bradley–Terry arena. We study a different operation, removing individual *evaluation items*, for the weighted-accuracy scores used in reward-model and judge meta-evaluation. For these scores the question reduces to a linear program with a closed-form lower bound, so we can measure how close curation comes to the bound. Our contributions are:

- A lower bound: overtaking the leader requires removing at least the score gap (Prop. 1).
- A measurement: on RewardBench and RewardBench 2, verified item subsets reach this bound near the top of the leaderboard, and stay within about twice it when each section keeps its share.
- A baseline without an adversary: random validation sets of realistic size often put a model other than the official leader first.

2 Setup

A benchmark has items $i = 1, \ldots, N$ with official weights $w_i^0 \geq 0$, $\sum_i w_i^0 = 1$. Model m has per-item score $r_{mi} \in [0, 1]$ and official score

$S_m = \sum_i w_i^0 r_{mi}$. A *curation* keeps a set K with kept share $f = \sum_{i \in K} w_i^0$ and rescores every model on it, $S_m^K = \sum_{i \in K} w_i^0 r_{mi} / f$. We call $1 - f$ the *removed share*.

Data. RewardBench v1 publishes per-item results for 118 models on 2985 preference pairs in 23 subsets (Lambert et al., 2024). We rebuild each official score from them: a section score is the count-weighted mean of its subset accuracies, with `math-prm` upweighted as in the released code, and the overall score is the mean of the four section scores. For four models, from the leader to a 0.5B chat model, our subset accuracies match RewardBench's published values exactly. Result files are aligned by position; a text check found no misaligned items. RewardBench 2 (Malik et al., 2025) publishes results by item id. We use the 188 models with results on all 1753 items of its five accuracy subsets and score them as the mean of subset accuracies. We exclude the Ties subset, which uses a different metric.

Crowning. A set K *crowns* model m if $S_m^K > S_k^K$ for every other model k. For a fixed kept share f, relaxing the subset to item weights w turns this into a linear program:

$$\max_{w,t} \; t \;\; \text{s.t.} \;\; \sum_i w_i (r_{mi} - r_{ki}) \geq t \;\; \forall k \neq m,$$
$$0 \leq w_i \leq w_i^0 / f, \quad \sum_i w_i = 1.$$

Model m is crownable at f if $t^* > 0$. The *section-preserving* variant also fixes each section's total weight at its official value, so the curated set keeps the benchmark's composition. We scan f downward from 1 and then bisect. Each LP solution is rounded to an item subset by keeping the items at or near their upper bound, each at its official weight, renormalised. The subset is rescored and counts only if m is still strictly first, so every reported share belongs to a verified subset. Because the search over f is heuristic, and any verified section-preserving subset is also valid without the constraint, we report the unconstrained cost as the smaller of the two. Each feasibility check is an LP with N weights and $M - 1$ constraints. By contrast, the analogous selection problem for ordinal aggregates such as mean win rate is NP-hard (Gordienko et al., 2026).

3 A Lower Bound on the Cost of Curation

Proposition 1. *If $S_a - S_b = g > 0$, then every K with $S_b^K \geq S_a^K$ removes a share of at least g.*

Proof. Let $D_i = r_{ai} - r_{bi} \in [-1, 1]$. Then $f\,(S_a^K - S_b^K) = \sum_{i \in K} w_i^0 D_i = g - \sum_{i \notin K} w_i^0 D_i \geq g - (1 - f)$, which is positive whenever $1 - f < g$. □

Crowning a model means overtaking the leader, so the removed share is at least that model's gap to the leader. For a single pair the bound is tight under a simple condition. Let $W_{ab} = \sum_i w_i^0 \, \mathbf{1}[r_{ai} = 1, \, r_{bi} = 0]$ be the weight of items that a gets right and b gets wrong. If $W_{ab} \geq g$, removing weight $g + \epsilon$ of these items makes b overtake a for any $\epsilon > 0$, so the pairwise cost equals g up to the weight of one item. On RewardBench the condition holds for 113 of the 117 models behind the leader. With many models the bound need not be reachable, because removing items that favour the leader can let a third model overtake both. Whether it is reached is therefore an empirical question.

4 Results

The bound is reached near the top. On RewardBench every one of the 118 models can be crowned. The ratio of removed share to score gap has a median of 1.00, and at most 1.02 in the top 20, so near the top the multi-model cost equals the pairwise bound (Figure 1, Table 1). The largest ratio over all models is 2.00. The leader is 0.7 points ahead of the runner-up, and removing 19 of 2985 items (0.7% of the score weight) reverses their order. Every top-20 model, down to one 10.5 points behind the leader, can be crowned while keeping at least 89.5% of the benchmark; 19 of the 118 models can be crowned keeping 90%, and 58 keeping 80%. RewardBench 2 gives the same picture: the ratio has a median of 1.00 and a maximum of 1.35, and its runner-up needs 5.3% removed against a 5.2-point gap.

Preserving composition raises the cost modestly. A curator who keeps every section at its official share leaves no trace in the per-section breakdown. On RewardBench this costs a median of 1.39 times the gap and at most 2.31, and every top-20 model can still be crowned keeping at least 81.8% of the benchmark. On RewardBench 2 the median is 1.12 and the maximum 1.58.

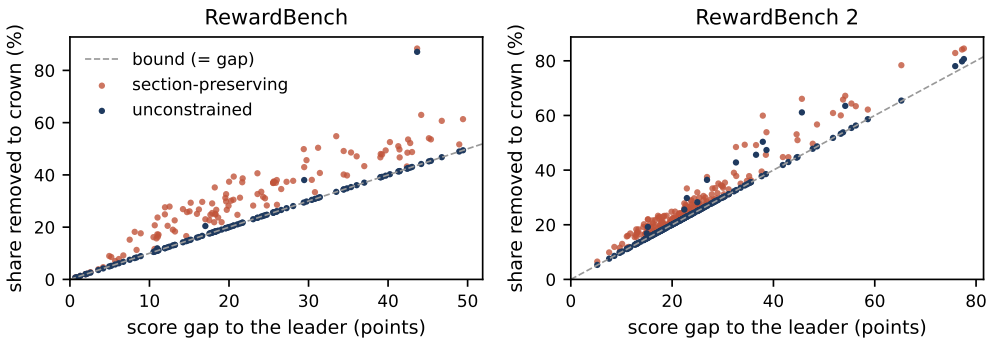


Figure 1: Share of the benchmark removed to put each model first, against its official gap to the leader. Every point lies on or above the dashed line $y = x$, the bound of Prop. 1. Unconstrained curation reaches the bound for nearly every model; keeping each section's share costs more, at most 2.31 times the gap.

			Removed to crown		
Model	Score	Gap	Share	Items	Strat.
QRM-Gemma-2-27B	94.4	0.67	0.68	19	0.72
Skywork-Reward-Gemma-2-27B-v0.2	94.3	0.84	0.85	30	0.85
Skywork-Reward-Gemma-2-27B	93.8	1.30	1.33	38	1.34
QRM-Llama3.1-8B-v2	93.1	1.97	1.98	61	1.98
Skywork-Reward-Llama-3.1-8B-v0.2	93.1	1.98	2.02	59	2.06
URM-LLaMa-3.1-8B	92.9	2.17	2.19	71	2.19
Skywork-Reward-Llama-3.1-8B	92.5	2.58	2.58	80	2.77

Table 1: The seven models behind RewardBench's leader: official score and gap to the leader (points), the share of score weight and the number of items whose removal crowns each model, and the share needed when every section keeps its official weight (Strat.).

	RewardBench		RewardBench 2	
n	leader kept	winners	leader kept	winners
25	14%	60	41%	113
50	25%	33	66%	64
100	35%	20	87%	18
200	48%	12	97%	5
500	65%	9	100%	2
1000	78%	6	100%	1

Table 2: Random validation sets of n items (2,000 draws each). "Leader kept": share of draws in which the official leader is strictly first after random tie-breaking. "Winners": number of distinct models that come first at least once.

Curation without an adversary. Table 2 draws random validation sets of n items, each item drawn with probability proportional to its official weight, so that a set's mean is an unbiased estimate of the official score. At $n = 100$, RewardBench's leader, 0.7 points ahead, comes first in 35% of draws, and 20 different models come first at least once. RewardBench 2's leader is 5.2 points ahead and comes first in 87% of 100-item draws. As Prop. 1 predicts, the size of the lead, not the benchmark, determines how stable the ranking is. Natural slices also change the winner: scoring on one section at a time gives 3 different winners across the four sections, and among single-subset slices with a unique winner, the donotanswer slice is won by the model ranked #95 overall.

5 Implications for Automated Pipelines

A pipeline that selects its evaluator on items it curated, or on a slice chosen after seeing results, should not trust a lead smaller than the share of items it could plausibly have dropped. Three practices follow. (i) Report the *curation margin* of a selection. By Prop. 1 it is at least the score gap, and near the top of both leaderboards it equals the gap, so RewardBench's 0.7-point lead is a 0.7% curation margin. (ii) Fix and hash the validation items before any candidate is scored, and log every subset or filter the pipeline applies. (iii) Size validation sets for rank stability, not only for accuracy. A hundred well-chosen items can estimate a model's accuracy closely (Maia Polo et al., 2024), but when the lead is under a point, as on RewardBench, the official leader comes first in fewer than half of the draws at $n \leq 200$ (Table 2), consistent with measured benchmark variance (Madaan et al., 2024).

Limitations

We study one curation operation, removing items, on two reward-model benchmarks. Reweighting the kept items, paraphrasing them or changing the prompt format would give a curator more freedom, so our costs overstate what curation needs. The LP relaxation is rounded and re-verified, so every reported share is achievable, but it may not be the

smallest possible. A model tied with another on every item could never be strictly first; no such pair occurred, and every model on both benchmarks was crownable. We do not claim that any published result was curated. RewardBench 2's Ties subset is excluded, so its scores differ slightly from the official ones.

Reporting and Reproducibility

All results come from public per-item result files (`allenai/reward-bench-results`, `allenai/reward-bench-2-results`). All analysis ran on one laptop CPU with SciPy's HiGHS solver; each full sweep takes one to two hours, and no models were run. One failed run is worth reporting: the first section-preserving search began its bisection at a very small kept share, where rounding can empty a section, and wrongly reported the leader as not crownable. The corrected search scans from $f = 1$ downward. Code and derived data will be released with the camera-ready version.

Use of AI Assistants

An AI assistant (Claude, Anthropic) helped write the analysis code and draft the text. The author chose the study, checked the proof and the results, and takes responsibility for the content.

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